

Table of Contents

RecommendedMetTriggers2017	1
RECOMMENDEDRIGGERS 2017	1
RECOMMENDEDRIGGERS 2018	5
HT.....	5

RecommendedMetTriggers2017

RECOMMENDED TRIGGERS 2017

Trigger Group	Single jet	Multi jets
period B (all runs except 325789, 327761, 328017)	small-R HLT_j380 (+) HLT_j400 HLT_j225_gsc380_boffperf_split (+) HLT_j225_gsc400_boffperf_split large-R trimmed HLT_j420_a10t_lcw_jes_L1J100 (+) (++) HLT_j440_a10t_lcw_jes_L1J100 (++) HLT_j390_a10t_lcw_jes_30smcINF_L1J100 (++) large-R untrimmed HLT_j420_a10_lcw_subjes_L1J100 (+) HLT_j440_a10_lcw_subjes_L1J100 large-R reclustered HLT_j420_a10r_L1J100 (+) HLT_j440_a10r_L1J100	small-R HLT_3j200 HLT_4j100 (+) 4j120 HLT_4j60_gsc100_boffperf_split (+) HLT_4j60_gsc115_boffperf_split HLT_5j70_L14J15 HLT_5j50_gsc70_boffperf_split (L14J20) HLT_5j50_gsc70_boffperf_split_L14J150ETA2 HLT_6j60_L14J15 HLT_6j45_gsc60_boffperf_split (L14J20) HLT_6j45_gsc60_boffperf_split_L14J150ETA2 HLT_7j45_L14J15 HLT_7j25_gsc45_boffperf_split_L14J20 HLT_7j25_gsc45_boffperf_split_L14J150ETA2 HLT_10j40_L14J15 small-R central HLT_5j65_0eta240_L14J15 HLT_5j50_gsc65_boffperf_split_0eta240 (L14J20) HLT_5j50_gsc65_boffperf_split_0eta240_L14J15 HLT_6j45_0eta240 (L14J15) (+) HLT_6j45_0eta240_L14J20) (++) HLT_6j50_0eta240_L14J15 HLT_6j25_gsc45_boffperf_split_0eta240_L14J20 HLT_6j25_gsc45_boffperf_split_0eta240_L14J15 (+) HLT_6j25_gsc50_boffperf_split_0eta240_L14J20 HLT_6j25_gsc50_boffperf_split_0eta240_L14J15 large-R trimmed HLT_2j330_a10t_lcw_jes_30smcINF_L1J100 (+)
period B (all runs except 325789)	small-R HLT_j420 HLT_j225_gsc420_boffperf_split large-R trimmed HLT_j460_a10t_lcw_jes_L1J100 HLT_j420_a10t_lcw_jes_40smcINF_L1J100	small-R HLT_3j200 HLT_4j120 HLT_4j60_gsc115_boffperf_split HLT_5j85_L14J15 HLT_5j60_gsc85_boffperf_split (L14J20) HLT_5j60_gsc85_boffperf_split_L14J150ETA2 HLT_6j70_L14J15

Trigger Group	Single jet	Multi jets
	large-R untrimmed HLT_j460_a10_lcw_subjes_L1J100 large-R reclustered HLT_j460_a10r_L1J100	HLT_6j50_gsc70_boffperf_split (L14J20) HLT_6j50_gsc70_boffperf_split_L14J150ETA2 HLT_7j45_L14J15 HLT_7j25_gsc45_boffperf_split_L14J20 HLT_7j25_gsc45_boffperf_split_L14J150ETA2 HLT_10j40_L14J15 small-R central HLT_5j70_0eta240_L14J15 HLT_5j50_gsc70_boffperf_split_0eta240 (L14J20) HLT_5j50_gsc70_boffperf_split_0eta240_L14J150ETA25 HLT_6j55_0eta240_L14J15 HLT_6j35_gsc55_boffperf_split_0eta240_L14J20 HLT_6j35_gsc55_boffperf_split_0eta240_L15J1 large-R trimmed HLT_2j330_a10t_lcw_jes_30smcINF_L1J100 (-)
period B (all runs)	small-R HLT_j450 HLT_j225_gsc440_boffperf_split large-R trimmed HLT_j480_a10t_lcw_jes_L1J100 HLT_j440_a10t_lcw_jes_40smcINF_L1J100 large-R untrimmed HLT_j480_a10_lcw_subjes_L1J100 large-R reclustered HLT_j480_a10r_L1J100	small-R HLT_3j225 HLT_4j120 HLT_4j60_gsc115_boffperf_split HLT_5j90 (L14J20) HLT_5j90_L14J150ETA25 HLT_5j60_gsc90_boffperf_split (L14J20) HLT_5j60_gsc90_boffperf_split_L14J150ETA2 HLT_6j70 (L14J20) HLT_6j50_gsc70_boffperf_split (L14J20) HLT_6j50_gsc70_boffperf_split_L14J150ETA2 HLT_7j45 (L16J15) HLT_7j45_L14J150ETA25 HLT_7j45_L14J20 HLT_7j25_gsc45_boffperf_split_L14J20 HLT_7j25_gsc45_boffperf_split_L14J150ETA2 HLT_10j40_L14J150ETA25 HLT_10j40_L14J20 HLT_10j40_L16J15 small-R central HLT_5j75_0eta240 (L14J20) HLT_5j75_0eta240_L14J150ETA25 HLT_5j55_gsc75_boffperf_split_0eta240 (L14J20) HLT_5j55_gsc75_boffperf_split_0eta240_L14J150ETA25 HLT_6j60_0eta240_L14J20 HLT_6j60_0eta240_L14J150ETA25 HLT_6j35_gsc55_boffperf_split_0eta240_L14J20 HLT_6j35_gsc55_boffperf_split_0eta240_L15J1 large-R trimmed HLT_2j330_a10t_lcw_jes_40smcINF_L1J100
period C	small-R HLT_j400 (+++)	small-R HLT_3j200

Trigger Group	Single jet	Multi jets
	HLT_j420 HLT_j225_gsc400_boffperf_split (+++) HLT_j225_gsc420_boffperf_split large-R trimmed HLT_j440_a10t_lcw_jes_L1J100 (+++) HLT_j460_a10t_lcw_jes_L1J100 HLT_j390_a10t_lcw_jes_30smcINF_L1J100 (+++) HLT_j420_a10t_lcw_jes_40smcINF_L1J100 large-R untrimmed HLT_j440_a10_lcw_subjes_L1J100 (+++) HLT_j460_a10_lcw_subjes_L1J100 large-R reclustered HLT_j440_a10r_L1J100 (+++) HLT_j460_a10r_L1J100	HLT_4j120 HLT_4j60_gsc115_boffperf_split HLT_5j70_L14J15 (+++) HLT_5j85_L14J15 HLT_5j50_gsc70_boffperf_split (L14J20) (+++) HLT_5j50_gsc70_boffperf_split_L14J150ETA2 HLT_5j60_gsc85_boffperf_split (L14J20) HLT_5j60_gsc85_boffperf_split_L14J150ETA2 HLT_6j60_L14J15 (+++) HLT_6j70_L14J15 HLT_6j45_gsc60_boffperf_split (L14J20) (+++) HLT_6j45_gsc60_boffperf_split_L14J150ETA2 HLT_6j50_gsc70_boffperf_split (L14J20) HLT_6j50_gsc70_boffperf_split_L14J150ETA2 HLT_7j45_L14J15 HLT_7j25_gsc45_boffperf_split_L14J20 HLT_7j25_gsc45_boffperf_split_L14J150ETA2 HLT_10j40_L14J15 small-R central HLT_5j65_0eta240_L14J15 (+++) HLT_5j70_0eta240_L14J15 HLT_5j50_gsc65_boffperf_split_0eta240 (L14J20) (+++) HLT_5j50_gsc65_boffperf_split_0eta240_L14J15 (+++) HLT_5j50_gsc70_boffperf_split_0eta240 (L14J20) HLT_5j50_gsc70_boffperf_split_0eta240_L14J15 HLT_6j50_0eta240_L14J15 (+++) HLT_6j55_0eta240_L14J15 HLT_6j25_gsc50_boffperf_split_0eta240_L14J15 HLT_6j25_gsc50_boffperf_split_0eta240_L14J15 (+++) HLT_6j35_gsc55_boffperf_split_0eta240_L14J15 HLT_6j35_gsc55_boffperf_split_0eta240_L14J15 (+++) HLT_7j25_gsc45_boffperf_split_L14J150ETA2 large-R trimmed HLT_2j330_a10t_lcw_jes_30smcINF_L1J100 (-) HLT_2j330_a10t_lcw_jes_40smcINF_L1J100
period D1-D5	small-R HLT_j400 (+++) HLT_j420 HLT_j225_gsc400_boffperf_split (+++) HLT_j225_gsc420_boffperf_split large-R trimmed HLT_j440_a10t_lcw_jes_L1J100 (+++) HLT_j460_a10t_lcw_jes_L1J100 HLT_j390_a10t_lcw_jes_30smcINF_L1J100 (+++) HLT_j420_a10t_lcw_jes_40smcINF_L1J100	small-R HLT_3j200 HLT_4j120 HLT_4j60_gsc115_boffperf_split HLT_5j70_L14J15 (+++) HLT_5j85_L14J15 HLT_5j50_gsc70_boffperf_split (L14J20) (+++) HLT_5j50_gsc70_boffperf_split_L14J150ETA2 HLT_5j60_gsc85_boffperf_split (L14J20) HLT_5j60_gsc85_boffperf_split_L14J150ETA2 HLT_6j60_L14J15 (+++) HLT_6j70_L14J15

Trigger Group	Single jet	Multi jets
	HLT_j420_a10t_lcw_jes_40smcINF_L1J100 large-R untrimmed HLT_j440_a10_lcw_subjes_L1J100 (+++) HLT_j460_a10_lcw_subjes_L1J100 large-R reclustered HLT_j440_a10r_L1J100 (+++) HLT_j460_a10r_L1J100	HLT_6j70_L14J15 HLT_6j45_gsc60_boffperf_split (L14J20) (+++) HLT_6j45_gsc60_boffperf_split_L14J150ETA2 HLT_6j50_gsc70_boffperf_split (L14J20) HLT_6j50_gsc70_boffperf_split_L14J150ETA2 HLT_7j45_L14J15 HLT_7j25_gsc45_boffperf_split_L14J20 HLT_7j25_gsc45_boffperf_split_L14J150ETA2 HLT_10j40_L14J15 small-R central HLT_5j65_0eta240_L14J15 (+++) HLT_5j70_0eta240_L14J15 HLT_5j50_gsc65_boffperf_split_0eta240 (L14J20) (+++) HLT_5j50_gsc65_boffperf_split_0eta240_L14J150ETA2 (+++) HLT_5j50_gsc70_boffperf_split_0eta240 (L14J20) (+++) HLT_5j50_gsc70_boffperf_split_0eta240_L14J150ETA2 (+++) HLT_6j50_0eta240_L14J15 (+++) HLT_6j55_0eta240_L14J15 HLT_6j25_gsc50_boffperf_split_0eta240_L14J150ETA2 (+++) HLT_6j25_gsc50_boffperf_split_0eta240_L14J150ETA2 (+++) HLT_6j35_gsc55_boffperf_split_0eta240_L14J150ETA2 (+++) HLT_6j35_gsc55_boffperf_split_0eta240_L14J150ETA2 (+++) large-R trimmed HLT_2j330_a10t_lcw_jes_30smcINF_L1J100 (-) HLT_2j330_a10t_lcw_jes_40smcINF_L1J100
period D6-K		

(*) MET trigger option that gives you the best combination of low thresholds & small loss of luminosity (~270 pb⁻¹ loss of luminosity w.r.t. option ***)

(**) MET trigger option that gives you lower thresholds (L1_XE50) but has been prescaled for part of the runs (~ 900 pb⁻¹ loss of luminosity w.r.t. option ***)

(***) MET trigger option that gives you full luminosity at the cost of a higher threshold (recommended if you prefer not to combine triggers over different data-taking periods)

(+) Jet trigger also disabled in runs 327582, 327745, 327860, 328221 and 328374 of period B (~1.2ifb loss)

(++) Jet trigger slightly different combination of missing runs / part-enabling of trigger for different parts of some runs

(+++) Jet trigger missing small fraction of luminosity from periods C (0.14 from 2.77 ifb) and D1-D5 (0.35 from 5.90 ifb)

Important remarks:

- The above quoted luminosity numbers are only indicative. Always assess the luminosity yourself to weigh threshold vs. luminosity arguments for your analysis.
- Further to this, please see JetTriggersToUseIn2017 coupled with the 1.7 and 1.5e34 menus from L34TriggerMenu2017 for the jet trigger group's recommendations and the available options.
- Before run 332655 (the deployment of release 21.1.10) jet GSC triggers were off for a few lumi blocks just after the start of the run while the beamspot determination was under way. A special GRL is provided for analysers using these triggers, removing 262 of the 88487 lumi blocks, corresponding to 73 pb⁻¹ out of the total 40404.4 pb⁻¹ in the data17_13TeV.periodAllYear_DetStatus-v97-pro21-13_Unknown_PHYS_StandardGRL_All_Good_25ns.xml GRL
- Analyses using MET triggers should also check events that fire **HLT_noalg_L1J400** to see if any of these events were **not** fired by the met trigger in the analysis phase-space.

RECOMMENDED TRIGGERS 2018

Trigger Group	MET
period B	HLT_xe110_pufit_xe70_L1XE50 HLT_xe120_pufit_L1XE50
period C-J	HLT_xe110_pufit_xe65_L1XE50* HLT_xe110_pufit_xe70_L1XE50 HLT_xe120_pufit_L1XE50
period K-	HLT_xe110_pufit_xe65_L1XE50 HLT_xe110_pufit_xe70_L1XE50

* Unprescaled since run 350067

HT

Trigger Group	HT
period B-	HLT_ht1000_L1J100

Major updates:

-- MoritzBackes - 2017-08-09

Responsible: MoritzBackes

Last reviewed by: **Never reviewed**

This topic: Atlas > RecommendedMetTriggers2017

Topic revision: r2 - 2023-03-20 - HollyPacey



Copyright & © 2008-2026 by the contributing authors. All material on this collaboration platform is the property of the contributing authors.

or Ideas, requests, problems regarding TWiki? use Discourse or Send feedback

```
var _paq = window._paq = window._paq || []; /* tracker methods like
"setCustomDimension" should be called before "trackPageView" */
_paq.push(['trackPageView']); _paq.push(['enableLinkTracking']);
(function() { var u="https://webanalytics.web.cern.ch/";
_paq.push(['setTrackerUrl', u+'matomo.php']); _paq.push(['setSiteId',
'5']); var d=document, g=d.createElement('script'),
s=d.getElementsByTagName('script')[0]; g.async=true; g.src=u+'matomo.js';
s.parentNode.insertBefore(g,s); })();
```